Rithwik Veera Balaji Kumar






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Professional Summary

Software Engineer with 1+ years of experience designing and building cloud-native, full-stack applications using **React.js**, **Node.js**, **Python**, and **AWS**. Delivered secure, scalable backend services, **RESTful APIs**, and event-driven serverless systems on **AWS**. Built **agentic AI workflows** and **RAG**-based conversational interfaces using **LangChain** and **Amazon Bedrock**. Growth-focused professional with strong customer-facing skills, eager to deliver both scalable software and impactful AI solutions.

Education

Georgia Institute of Technology **Jan 2026 - Jan 2028**
M.S. in Computer Science, Specialization in Artificial Intelligence (Online) *GPA: 4.0*

National Institute of Technology Tiruchirappalli **Jul. 2019 – Jul. 2023**
B.Tech in Instrumentation and Control Engineering, Minor in Computer Applications *CGPA: 8.34*

Experience

Amazon Web Services **Aug 2023 – Oct 2024**
Application Developer *Bangalore, KA*

- Combined cloud-native application engineering with customer-facing consulting within **Professional Services**, delivering user stories across **Agile/Scrum** sprints with **cross-functional** customer teams to translate evolving business needs into solutions.
- Contributed **frontend development** for a greenfield engagement, building **production-grade**, **user-facing** tools and **API integrations** with **React.js**, **Node.js**, **Material UI**, and **Bootstrap**, boosting satisfaction by **50%**.
- Deployed secure serverless backend infrastructure using **AWS Amplify**, **AppSync (GraphQL)**, **Lambda**, **IAM**, **Cognito**, and **WAF**, reducing customers' manual operational efforts **by 70%** while improving access control and reliability **by 40%**.
- Developed **Generative AI** solutions on **AWS Bedrock** using **LangChain**, applying prompt engineering and knowledge-base integration, and employing **RAG** to deliver production-ready conversational interfaces for customer use cases.

Beam Global Spirits & Wine (India) Pvt. Ltd. **June 2022 – July 2022**
Data Engineering Intern *Gurgaon, HR*

- Crafted a Product Mapping solution using **Python** & **Pandas** to standardize datasets and improve catalog accuracy.
- Optimized matching logic using libraries like **FuzzyWuzzy**, **regex**, and **string_grouper** (TF-IDF + cosine similarity), increasing product classification and entity resolution accuracy to **85%+**.

Sri Mookambika Infosolutions Pvt Ltd. **March 2020 – July 2020**
Summer Intern *Madurai, TN*

- Built a **Django-based student portal** with secure registration and backend logic for smooth data handling.
- Designed a responsive UI using **React Bootstrap** and admin **CRUD**, enhancing UX and improving control **by 50%**.

Projects

LangChain Documentation Helper | *LangChain, RAG, Claude, Pinecone, Ollama, LangSmith, RAGAs* **Apr 2025 – Present**

- Architected a **Retrieval-Augmented Generation (RAG)** documentation assistant using **Claude**, **Ollama**, and **Pinecone** to answer questions from LangChain's own documentation with cited sources, eliminating hallucinated answers.
- Achieved complete, deterministic documentation coverage across LangChain, LangGraph, Deep Agents, and LangSmith docs via a **sitemap-crawl ingestion pipeline** generating **Ollama embeddings** into a **Pinecone vector database**.
- Improved answer precision and **domain enforcement** by pairing confidence-scored **semantic retrieval** with a **query-rewrite** step that resolves multi-turn follow-ups before retrieval.
- Achieved **94% faithfulness** and **89% answer relevance** on a 50-example golden dataset via a custom **6-evaluator RAGAs/LangSmith** evaluation framework with full request-level **tracing** for production **observability** and **debugging**.
- Delivered a responsive chat experience via a **Streamlit** interface with token-by-token streaming and inline source citations.

Console Companion | *Amazon Bedrock, AWS Config, Streamlit, Python* **Jul 2024 – Oct 2024**

- Created a **GenAI chatbot** using **Bedrock**, **Streamlit**, and **AWS SDK for Config** to help users track multi-account and multi-region AWS resources, addressing compliance validation and audit challenges.
- Orchestrated an **agentic AI workflow** using **Bedrock Agents**, defining Lambda tools via custom, reusable **OpenAPI action-group schemas** that define the agent-to-tool contract for **ReAct**-style reasoning.
- Built a **Validator Agent** with **AWS Lambda** and **LangChain** that validates generated SQL against custom compliance rules and execution errors to refine generation, improving **agent reliability** by **80%**.
- Engineered **RAG** using **Bedrock KBs** over resource-type-partitioned datasets via a **Python-based ETL pipeline** into **AWS S3**, raising **context quality** to cut token usage and latency while improving SQL generation accuracy to **90%**.
- Formulated an LLM chat UI with **Streamlit** to surface the **agentic workflow**, improving interaction flow **by 40%**.

Car Sharing Application (AWS Tech U Capstone) | *React, GraphQL, AWS* **Nov 2023 – Jan 2024**

- Programmed a cloud-based car-sharing app with **React.js** and **GraphQL**, featuring user booking and ride tracking.
- Leveraged **Amplify** & **AppSync** to optimize the backend, cutting query time by **30%** and dev effort by **60%**.
- Integrated **Cognito** user pools and **IAM** roles to enable secure user authentication and access authorization.
- Employed **AWS Lambda** for ride processing and **DynamoDB** for user storage, reducing system latency by **40%**.
- Deployed app via **WAF**, **Route53**, & **CloudFront**, boosting traffic handling and delivery speeds by **50%**.

Certifications

AWS Certified Solutions Architect Associate – *Amazon Web Services Training and Certification*

AWS Certified AI Practitioner – *Amazon Web Services Training and Certification*

Terraform Associate – *HashiCorp*

AWS Certified Cloud Practitioner – *Amazon Web Services Training and Certification*

Skills

Programming Languages: Python, C/C++, Java, JavaScript, TypeScript

AI & Data: LLM-based Application Development, Agentic AI, AI Agents, RAG, ReAct, Reliability Patterns (Reflection, Reflexion, Validator Loops), Conversational AI, LangChain, LangGraph, LangSmith, SQL, Pandas, NumPy, PyTorch, Scikit-learn, ETL Pipelines

Full-Stack Development: HTML, CSS, ReactJs, NodeJs, Django, Streamlit, MaterialUI, Bootstrap, Testing, SDLC, Agile/Scrum

Cloud & DevOps: AWS (EC2, S3, Lambda, IAM, AppSync, Amplify, RDS, DynamoDB (NoSQL), API Gateway (REST), Cognito), Amazon Bedrock, Amazon Q, Terraform, CloudFormation, Microservices, Git, boto3, CI/CD